# PAKISTAN SPACE POLICY




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# Acknowledgement

*Space is the final frontier which has yet to be explored in depth by the Islamic Republic of Pakistan. In order to bring our nation in to the modern era of Nations who are exploring the Space and using it for the purpose of Atmospheric and Climate change Monitoring, Weather Forecasting, Defensive Purposes and to explore the deep space for the sake of knowledge and betterment of mankind, the Government of Pakistan wishes to state their Space Policy. The stance of the United Nations on 1967 Outer Space Treaty; the 1968 Rescue Agreement; the 1972 Liability Convention; the 1975 Registration Convention; and the 1979 Moon Agreement, is understood and reciprocated by the State of Pakistan.*

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## Space and the National Interest:

It is essential for the national interest that Pakistan maintains a robust, technologically superior and commercially competitive space industry. Few endeavors capture the imagination or tax the capabilities of humankind like the exploration and exploitation of space. And few have so quickly made themselves so indispensable.

## Government Action:

Space has been the focus of the State of Pakistan for quite some time and large strides have been taken by the Nation. For this Pakistan has taken some measures such as;

### SUPARCO:

Pakistan Space and Upper Atmosphere Research Commission (SUPARCO), the national space agency, was established in 1961 as a committee and was granted the status of a Commission in 1981. SUPARCO is mandated to conduct R&D in space science, space technology, and their peaceful applications in the country.

### Satellites:

The Pakistan Remote Sensing Satellite, commercially known as Remote Sensing Satellite System, is a dual-purpose Earth observational and optical satellite. Pakistan Remote Sensing Satellite-1 was launched from China's Jaquan Satellite Centre on 9 July 2018.

#### iCube-1:

It is a miniaturized satellite built by the Educational Sector, Institute of Space Technology Pakistan, with an objective to provide a wide range of future experiments in the domains of imaging, microgravity, space dynamic, and various fields. It was launched from Dombarovsky, Russia, on board the Dnepr launch vehicle.

### Space Education:

SUPARCO, the National Space Agency of Pakistan, designed and developed a robust and visionary programme titled as ‘Space Education and Awareness Drive (SEAD)’ aimed at Stimulating the next generation of space-oriented scientists, broadening young minds and help building the scientific and technological capacity of the country

The programme strives to not only familiarize children and young adults with current research, technology trends, and uses of space sciences and technology along-with the numerous career opportunities offered by the space sector but also to let them know that space science can be fun and inspiring.

### Launch Sites:

Tilla Satellite Launch Centre, primary known as Tilla Range, is a spaceport 25 kilometers west of Jhelum city in Punjab. It is also known as Mashhood Test Firing Range (MTFR). The spaceport has been used as a launch site for the **Hatf** and **Ghauri** ballistic missiles under development by the Khan Research Laboratories.

## Pakistan's Space Policy Framework:

| | PAKISTAN'S SPACE POLICY FRAMEWORK | | |
|---|---|---|---|
| **Principles** | 1. Pakistan's Interest First<br>2. Positioning the Private Sector at the Forefront of Space Activities<br>3. Progress Through Partnerships<br>4. Excellence in Key Capabilities<br>5. Inspiring Pakistanis | | |
| **Areas for Action** | **Commercialization** | **Research and Development** | **Exploration of Space** |
| | Stewardship, Management & Accountability | | |

## Five Core Principles:

### 1. Prioritizing National Interest:

National sovereignty, security and prosperity will be the key drivers of Pakistan's activities in space. Pakistan's first priority must be to use space effectively in support of these interests.

### 2.Positioning the Private Sector at the Forefront:

As space yields ever more commercial opportunities, the Government will focus on:

- Supporting the domestic space industry in the innovation required to bring to market cutting-edge technologies that meet national interests; and
- Utilizing industry where it has greater capacity, knowledge and skill, potential or when it can be more efficient and cost-effective.

### 3.Progress Through Partnerships:

Space is a shared domain and an expensive undertaking. The Government will look to continue partnerships to share the expenses and rewards of major space initiatives. This will include collaboration with international partners for mutual benefits and obtaining services and technologies that could otherwise be unavailable. At the same time, effective export control and regulatory measures will continue to protect Pakistani technologies. Effective Cybersecurity framework development will protect data from theft or from falling into the hands of hostile interests.

### 4.Excellence in Key Capabilities:

Pakistanis have had enormous success in a number of areas of space technology, from telecommunications to remote sensing to robotics – expertise that has been invaluable domestically and to our international partners. The Government will continue to support and

advance proven Pakistani competencies while keeping a close watch on new niches of technological accomplishment.

### 5.Inspiring Pakistanis:

An advanced, prosperous nation requires an educated, skilled workforce. Space is a highly visible means of motivating young Pakistanis to pursue careers in science, technology, engineering and math. Working with industries, universities and colleges, the Government will use every mean of communication to spread the message of the importance of space, in the effort to recruit, support and retain highly qualified and skilled personnel.

## Implementation

### 1.Commercialization:

Government has initiated many plans and projects for the public good in the areas, such as public safety, national defense, weather forecasting, environmental monitoring and disaster management. The Government will continue to ensure that it has access to the essential information and services it requires. At the same time, it commits to:

- Using the private sector, wherever feasible, to provide the equipment and services it needs;
- Providing the support to ensure that the domestic space industry is robust and globally competitive, especially by assisting in efforts to test and prove the value of new technologies;
- Pursuing consistent business models tailored to areas of activity from research and development (R&D) to space operations and a level of predictability and transparency that industry needs in order to make sound, strategic investment decisions; and
- Working to negotiate international agreements that open market access opportunities for Pakistani firms.

### 2.Research and Development:

As a high-technology enterprise, the lifeblood of the space industry is innovation, which in turn rests on research and development. Working with industry and the Pakistani space research community, the Government of Pakistan will encourage further opportunities in R&D and innovation by:

- Increasing support for technology development, especially in areas of proven strength such as robotics, optics, satellite communications and space-based radar, as well as in areas of emerging expertise;
- Coordinating with the granting councils and foundations to ensure that space research resources are leveraged and that space research itself figures prominently in their mandates; and

• Leveraging existing expertise and programs at the Pakistan Counsel of Scientific and Industrial Research (PCSIR), Ministry of Defense Production, Pakistan Centre for Development Communication and Pakistan Aerospace Counsel – including the Pakistan Counsel of Science and Technology – to better support the space industry

### 3. Exploration of Space:

Space exploration has changed our understanding of the universe and driven advances in science and leading-edge technologies. The spectacular success of the multi-nation collaboration that built and now operates the PRSS-1 – an engineering wonder – heralds a new era for space-based science. The Government commits to:

• Ensuring that Pakistanis a sought-after partner in the international space exploration missions that serve Pakistan's national interests;

• Continuing to invest in the development of Pakistani contributions in the form of advanced systems and scientific instruments as part of major international endeavors;

• Continuing Pakistan's Astronaut Program so as to have Pakistanis aboard current and future space laboratories and research facilities.

### 4. Stewardship, Management and Accountability:

Pakistan's commitments and initiatives in space must not be piecemeal. They have to be part of coordinated policies and strategies. In order to consult all pertinent partners in setting its future priorities, the Government will:

• Establish a Pakistani Space Advisory Council, representing the full range of stakeholders in the public and private space domain, chaired by the President of the Pakistani Space Agency.

At the same time, the Government will empower a committee chaired by a Deputy Minister to review objectives and expenditures.

## Civil Laws:

- Every citizen has equal right in contributing towards the space sector of Pakistan
- The private commercial sector should prioritize national interests
- Government and military shall have no precedence over private sector
- A request should be sent for the information about the materials to be exported to the "Pakistan Space Advisory Council" and then after the approval of the request items can be exported.
- In Government funded projects, material importing heavily depends on whether the material making is insufficient in the state.
- Monthly progress and accountability report should be sent to "Pakistan Space Advisory Council" for maintaining the records of all ongoing projects

- To ensure that every National (both private and Governmental) Space Program promotes space-related education and hold open-days for general public for public interests on National Holidays
- Frequent Seminars and workshops shall be held by Government and private sector to increase public engagement
- Quota system is to be abolished in Space Sector, only on talent and skills shall jobs be given.

## Future Trajectory

This policy framework creates Pakistan's space program for the third decade of the 21st century and beyond. It provides clarity with respect to Government priorities and strengthens governance of the space program. Implementation of its core principles will ensure that the Pakistani Space Program will be able to deliver the services the Government requires to protect and advance national interests; that industry will find the necessary support to be competitive in global markets; and that academia will be well positioned to conduct the research that will be the foundation of future space missions.

Space remains a harsh and unforgiving environment that tests the capabilities of even the most advanced technologies, spurring the development of ever more sophisticated engineering designs. Our use and understanding of space allow access to the most profound understanding of the cosmos and provides a platform for science to investigate the complexity of Earth as a living planet, even as space becomes increasingly indispensable to the well-being and security of life on Earth. Space is now a growing, multi-billion-dollar arena of commercial opportunity. It is imperative, then, that Pakistan remain in the vanguard of space research and application. This framework provides the policy blueprint for Pakistan to do so.

## Five-year Program:

- To promote Tilla range launch station to the rank of satellite launching base
- To create a Pakistani defense rocket
- To launch the first manned spacecraft from Pakistan
- To land the first Pakistani on Moon
- To collaborate with allied nations of Pakistan in order to send aa deep space exploratory mission
- To introduce space studies from matriculation level
- To send a communication satellite in collaboration with international organization
- To support National Space Awareness & Development Organizations in introducing innovative and state-of-the-art infrastructure for research-oriented outputs
- To assist NASA in its endeavor of launching a new Space Station
- To commercialize the space Flight